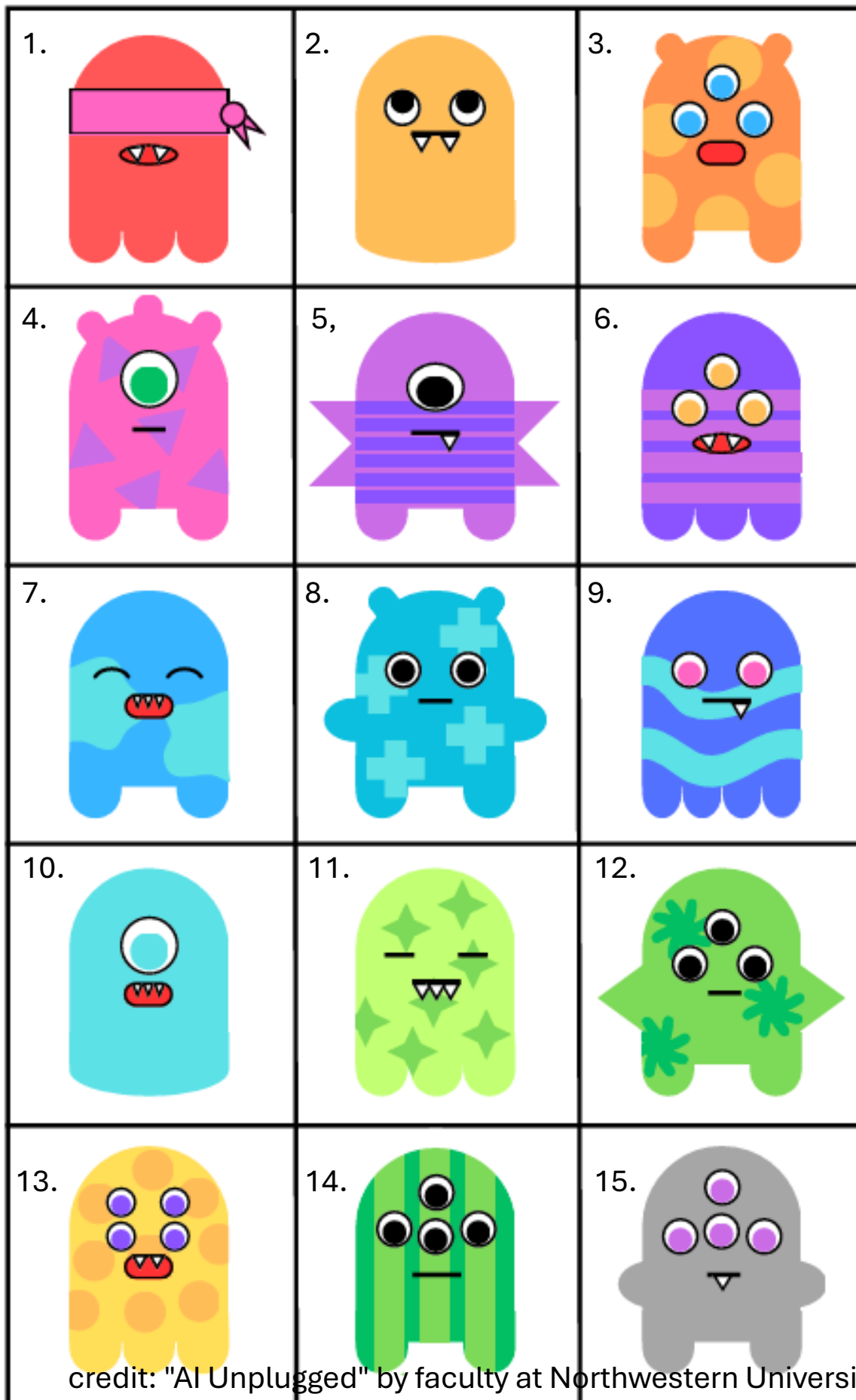


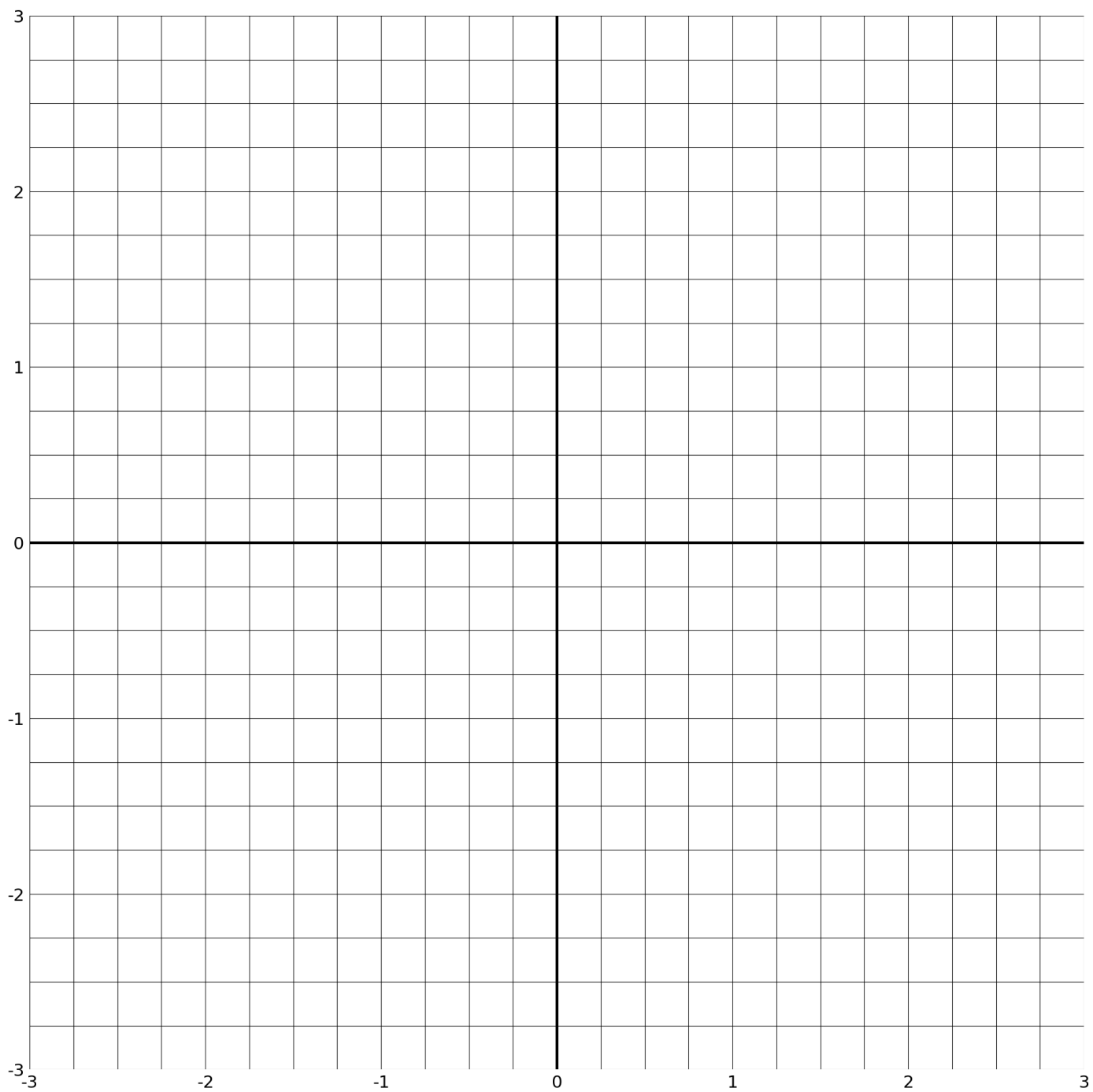


credit: "AI Unplugged" by faculty at Northwestern University

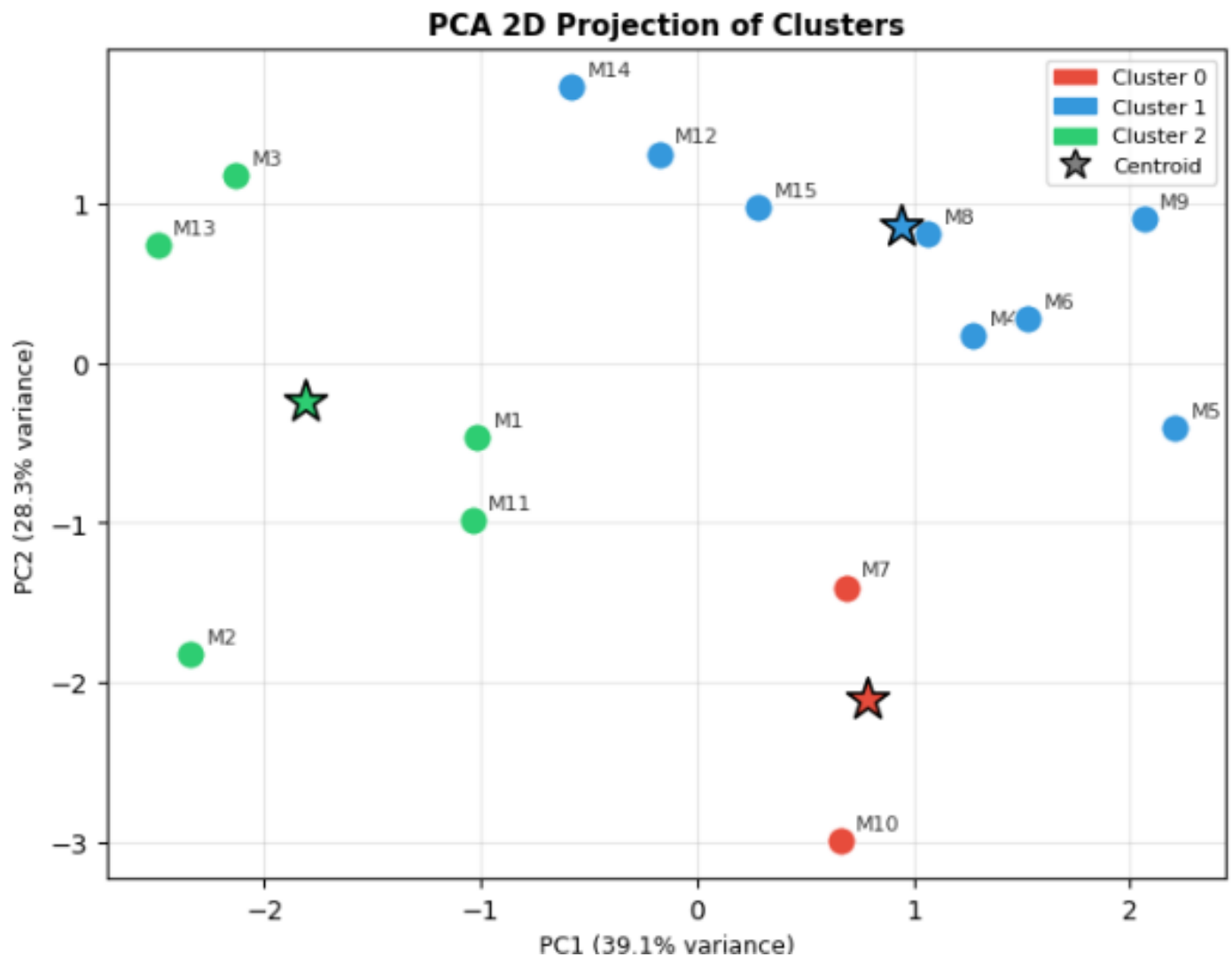
Monster Data:

MonsterID	Warmth	Eyes	Feet	Teeth	Solid	Wings
M1	1	2	3	2	1	0
M2	2	2	0	2	1	0
M3	2	3	2	0	0	0
M4	-1	1	2	0	0	0
M5	-2	1	2	1	0	1
M6	-2	3	3	2	0	0
M7	-1	2	2	3	0	0
M8	-1	2	2	0	0	1
M9	-2	2	4	1	0	0
M10	-1	1	0	3	1	0
M11	1	2	3	3	0	0
M12	0	3	2	0	0	1
M13	2	4	3	2	0	0
M14	0	4	2	0	0	0
M15	-1	4	2	1	1	1

Note: Warmth is a measure of Cool (-2) to Warm (+2) using the monster's primary coloration. Wings is a "yes" or "no" to whether the monster had wings and/or appendages. Yes = 1, No = 0
Solid is also Yes = 1 or No = 0. It records whether the monster was a solid color.



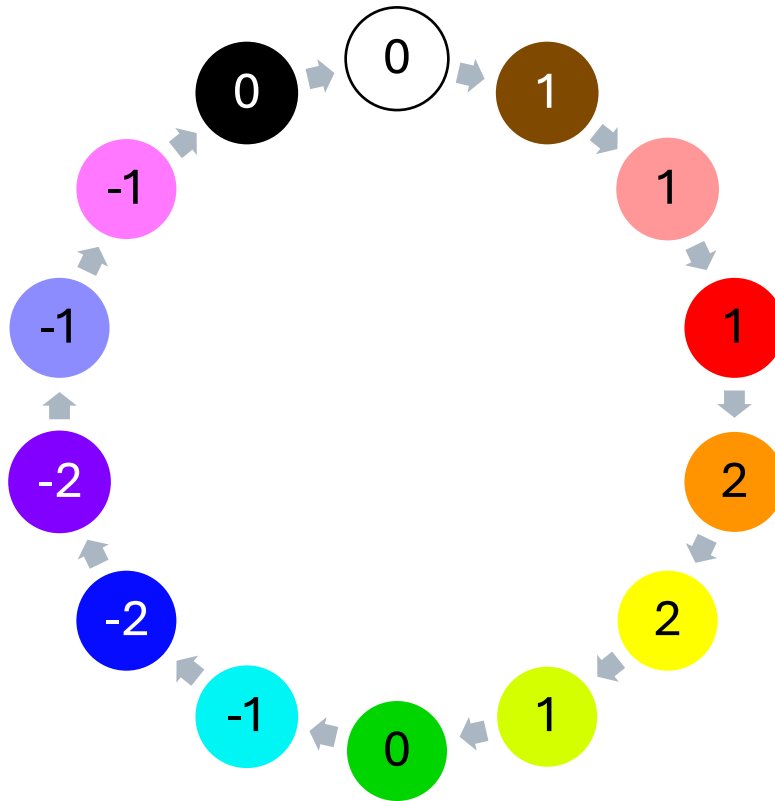
ACTUAL Solution Shown Here:



This is a plot of the **training dataset** with $k = 3$ and letting the algorithm work to find the best combination of variables as well as weights on them.

Interestingly, the same monsters ended up in the same clusters. Only their coordinates are slightly different because of the formulas based on maximizing the effect sizes.

This is how color "warmth" was selected:



PART 2: Testing the Model

Test Monster Data:

Note: "Averages" were calculated *from the training data*. We need to test the original model first. Then, if we have reason to make adjustments to the model, we do so. But not before.

	Warmth	Eyes	Feet	Teeth	Solid	Wings			Warmth	Eyes	Feet	Teeth	Solid	Wings
TestM1	-1	2	1	2	0	1		TestM1	-0.80	-0.40	-1.13	0.67	-0.27	0.73
TestM2	-1	2	0	3	1	1		TestM2	-0.80	-0.40	-2.13	1.67	0.73	0.73
TestM3	-2	3	3	0	1	1		TestM3	-1.80	0.60	0.87	-1.33	0.73	0.73
TestM4	2	2	2	1	1	0		TestM4	2.20	-0.40	-0.13	-0.33	0.73	-0.27
TestM5	2	3	3	2	1	1		TestM5	2.20	0.60	0.87	0.67	0.73	0.73
TestM6	2	2	2	2	1	1		TestM6	2.20	-0.40	-0.13	0.67	0.73	0.73
TestM7	-1	2	1	1	1	1		TestM7	-0.80	-0.40	-1.13	-0.33	0.73	0.73
TestM8	-1	2	4	0	1	1		TestM8	-0.80	-0.40	1.87	-1.33	0.73	0.73
TestM9	-1	1	1	1	0	1		TestM9	-0.80	-1.40	-1.13	-0.33	-0.27	0.73
TestM10	-2	2	0	0	1	1		TestM10	-1.80	-0.40	-2.13	-1.33	0.73	0.73
TestM11	0	2	3	0	0	0		TestM11	0.20	-0.40	0.87	-1.33	-0.27	-0.27
TestM12	0	3	3	4	1	1		TestM12	0.20	0.60	0.87	2.67	0.73	0.73
TestM13	1	3	2	4	1	1		TestM13	1.20	0.60	-0.13	2.67	0.73	0.73
TestM14	1	3	3	2	1	0		TestM14	1.20	0.60	0.87	0.67	0.73	-0.27
TestM15	1	1	3	4	0	1		TestM15	1.20	-1.40	0.87	2.67	-0.27	0.73
TestM16	-1	1	0	3	0	0		TestM16	-0.80	-1.40	-2.13	1.67	-0.27	-0.27
TestM17	0	2	2	1	0	0		TestM17	0.20	-0.40	-0.13	-0.33	-0.27	-0.27
Averages:	-0.20	2.40	2.13	1.33	0.27	0.27		"Centered" Data - (Mean-Adjusted)						

	X	Coeff	Y	Coeff
TestM1	0.75	-0.9293	-1.04	-0.0155
TestM2	0.45	-0.3091	-2.36	0.5213
TestM3	1.75	0.1315	1.61	0.4463
TestM4	-1.99	-0.093	-0.23	-0.6975
TestM5	-2.16	-0.0711	0.15	-0.1745
TestM6	-1.98	0.0997	-0.82	0.1087
TestM7	0.77		-0.52	
TestM8	1.26		1.52	
TestM9	1.15		-0.86	
TestM10	1.66		-0.25	
TestM11	0.17		1.12	
TestM12	-0.48		-1.21	
TestM13	-1.55		-1.67	
TestM14	-1.33		0.06	
TestM15	-0.72		-2.10	
TestM16	0.73		-2.81	
TestM17	-0.06		-0.02	

"Coeff" are the numbers for PC1 and PC2 that get multiplied by the "Centered Data" for each monster.

These are the projected coordinates, the results of the principal component analysis.

The model chooses what variables matter, and to what extent. The larger in absolute value the coefficient, the more weight that variable has.

Is the model stable? This table shows the predicted (assigned) cluster

Monster	k = 3	k = 4	k = 5	Stable?
TestM1	2	2	2	YES
TestM2	1	1*	1*	YES
TestM3	2	2	2	YES
TestM4	3	3	3	YES
TestM5	3	3	3	YES
TestM6	3	3	3	YES
TestM7	2	2	2	YES
TestM8	2	2	2	YES
TestM9	2	2	2	YES
TestM10	2	2	2	YES
TestM11	2	2	2	YES
TestM12	3	3	3	YES
TestM13	3	3	3	YES
TestM14	3	3	3	YES
TestM15	3	1*	3	NO
TestM16	1	1	1*	YES
TestM17	2	2	2	YES

* (required a tie breaking rule)

Credits:

For the monster images, I must give credit to the "AI Unplugged" group at Northwestern University. <https://sites.northwestern.edu/aiunplugged/>

The lesson plan is my own.